GIRI's Tech Hub, Pune.

SQL Machine Test

Batch: June-24 to June-25

Date: 20/02/2026

Timing: 02:30 to 05:00 Pm

Instructions :

- **Solve any 15 query.**

1. Hotel Table :-

Field	Type	Null	Key	Default	Extra
hotel_id	INT	NO	PRI	NULL	
hotel_name	VARCHAR(100)	YES		NULL	
location	VARCHAR(100)	YES		NULL	
rating	INT	YES		NULL	

2. Room Table :-

Field	Type	Null	Key	Default	Extra
room_id	INT	NO	PRI	NULL	
room_number	INT	YES		NULL	
room_type	VARCHAR(50)	YES		NULL	
price	DECIMAL(10,2)	YES		NULL	
hotel_id	INT	YES	MUL	NULL	

3. Guest Table :-

Field	Type	Null	Key	Default	Extra
guest_id	INT	NO	PRI	NULL	
guest_name	VARCHAR(100)	YES		NULL	
contact_no	VARCHAR(15)	YES		NULL	
check_in	DATE	YES		NULL	
check_out	DATE	YES		NULL	
room_id	INT	YES	MUL	NULL	

Solve This Queries :-

1. **Write a SQL query to display hotel_id, hotel_name, total number of rooms, and average room price for each hotel.**
   Only show hotels where the average room price is greater than 2000.
2. **Write a SQL query to display guest_name, room_number, room_type, and hotel_name for guests who are staying in rooms where the price is greater than 3000.**
3. **Write a SQL query to display hotel_name and the maximum room price available in each hotel.**
4. **Write a SQL query to display hotel_name and total rooms for hotels that have more than 5 rooms.**
5. **Write a SQL query to display guest_name, hotel_name, check_in, and check_out details of guests who checked in between '2025-01-01' and '2025-01-31'.**
6. **Write a SQL query to find the hotel that generates the highest revenue.**
   Revenue is calculated as SUM(price) of all booked rooms.
7. **Write a SQL query to display guest_name and hotel_name for guests who stayed in the most expensive room across all hotels.**
8. **Write a SQL query to display hotel_name of hotels that currently have no guests.**
9. **Write a SQL query to display hotel_name and average age of guests staying in each hotel.**
   Only show hotels where the average age is greater than 30.
10. **Write a SQL query to display guest_name, hotel_name, and number of days stayed for guests who stayed more than 3 days.**
11. **Write a SQL query to display room_id, room_number, price, and hotel_name for rooms whose price is greater than the average room price of their respective hotel using subquery.**
12. **Write a SQL query to display guest_name and hotel_name for guests staying in the hotel that has generated the highest total revenue. Revenue is calculated as SUM(price) of all booked rooms.**
13. **Create a view named Premium_Guest_View that displays:**

guest_name , hotel_name , room_number , room_type , price

The view should include only guests staying in rooms where price is greater than 4000.

14. Create a view named LongStayHotels that displays:

    hotel_name , average_stay_days

    Only include hotels where the average stay duration of guests is more than 2 days.

15. Create a stored procedure named GetGuestsByHotel that takes hotel_id as input and displays:

    guest_name

    room_number

    check_in

    check_out

16. Create a stored procedure named TransferGuest that:

    Takes guest_id and new_room_id as input

    Checks whether the new room exists

    Updates the guest's room_id

    Returns success or error message

17. Create a user-defined function named CalculateStayDays that takes check_in and check_out dates as input and returns total number of days stayed.

18. Create a function named GetHotelRevenue that takes hotel_id as input and returns total revenue generated by that hotel.

19. Create a BEFORE INSERT trigger on the Room table that prevents insertion if the price is less than or equal to 0.

20. Assume Room table has an additional column: availability (VARCHAR(20)).

    Create an AFTER INSERT trigger on Guest table that:

    Automatically sets room availability to 'Occupied' when a guest is inserted

    Updates the respective room record.

-----ALL THE BEST-----